

# Radioactive Decay and Half-Life

Nuclear equations, radiation and half-life

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

## RULES AND DATA

Alpha  ${}^4_2\text{He}$ :  $A - 4, Z - 2$  · Beta  ${}^0_{-1}\text{e}$ :  $Z + 1$  · Positron  ${}^0_{+1}\text{e}$ :  $Z - 1$  · Gamma: no change

$N = N_0 \times (\frac{1}{2})^n$ ,  $n = t \div t_{1/2}$  ·  $t = t_{1/2} \times \ln(N_0/N) \div \ln 2$

Z: N 7 · O 8 · Ca 20 · Ni 28 · Xe 54 · Rn 86 · Th 90 · Pa 91 · U 92

Use the same time unit for  $t$  and  $t_{1/2}$ . Show units on every quantity.

## A. Understanding

1. Define (a) radioactivity (b) half-life. (c) Why can we predict the half-life of a large sample but not the moment when one particular nucleus will decay?

---

---

---

---

## B. Nuclei and nuclear equations

2. Give the numbers of protons and neutrons in: (a)  ${}^{235}_{92}\text{U}$  (b)  ${}^{60}_{27}\text{Co}$  (c)  ${}^{131}_{53}\text{I}$  (d)  ${}^{222}_{86}\text{Rn}$

3. Complete each equation: (a)  ${}^{226}_{88}\text{Ra} \longrightarrow ? + {}^4_2\text{He}$  (b)  ${}^{131}_{53}\text{I} \longrightarrow ? + {}^0_{-1}\text{e}$  (c)  ${}^{18}_9\text{F} \longrightarrow ? + {}^0_{+1}\text{e}$  (d)  ${}^{60}_{27}\text{Co} \longrightarrow ? + {}^0_{-1}\text{e}$  (e)  ${}^{239}_{94}\text{Pu} \longrightarrow ? + {}^4_2\text{He}$  (f)  ${}^{40}_{19}\text{K} \longrightarrow ? + {}^0_{-1}\text{e}$

4. Identify X: (a)  ${}^{234}_{90}\text{Th} \longrightarrow {}^{234}_{91}\text{Pa} + \text{X}$  (b)  ${}^{210}_{84}\text{Po} \longrightarrow {}^{206}_{82}\text{Pb} + \text{X}$  (c)  ${}^{11}_6\text{C} \longrightarrow {}^{11}_5\text{B} + \text{X}$

5. Uranium-238 ( ${}^{238}_{92}\text{U}$ ) decays by alpha emission, then beta, then beta again. Write the three equations and identify the final nucleus.

### C. Types of radiation

---

6. Compare alpha, beta and gamma radiation: what each one is, its charge, and what stops it. Explain why an alpha emitter is most dangerous inside the body.

### D. Half-life

---

7. A patient is given technetium-99m with an activity of 800 MBq ( $t_{1/2} = 6.01 \text{ h}$ ). What is its activity after 24.04 h?

8. (a) The activity of a sample falls from 1600 Bq to 400 Bq in 30.0 min. Find its half-life. (b) Radon-222 has  $t_{1/2} = 3.82$  d. What percentage of a sample remains after 10.0 d?

### E. Challenge

---

9. A sample of carbon from an ancient wooden bowl has a carbon-14 activity of 0.0850 Bq per gram of carbon. Carbon from fresh wood measured in the same laboratory gives 0.230 Bq per gram. Estimate the age of the bowl ( $t_{1/2}$  of carbon-14 = 5730 yr).